

Animation and Graphics

Android provides a variety of powerful APIs for applying animation to UI elements and drawing custom 2D and 3D graphics. The sections below provide an overview of the APIs and system capabilities available and help you decide with approach is best for your needs.

Animation

The Android framework provides two animation systems: property animation (introduced in Android 3.0) and view animation. Both animation systems are viable options, but the property animation system, in general, is the preferred method to use, because it is more flexible and offers more features. In addition to these two systems, you can utilize Drawable animation, which allows you to load drawable resources and display them one frame after another.

Property Animation

Introduced in Android 3.0 (API level 11), the property animation system lets you animate properties of any object, including ones that are not rendered to the screen. The system is extensible and lets you animate properties of custom types as well.

View Animation

View Animation is the older system and can only be used for Views. It is relatively easy to setup and offers enough capabilities to meet many application's needs.

Drawable Animation

Drawable animation involves displaying [Drawable](#) resources one after another, like a roll of film. This method of animation is useful if you want to animate things that are easier to represent with Drawable resources, such as a progression of bitmaps.

2D and 3D Graphics

When writing an application, it's important to consider exactly what your graphical demands will be. Varying graphical tasks are best accomplished with varying techniques. For example, graphics and animations for a rather static application should be implemented much differently than graphics and animations for an interactive game. Here, we'll discuss a few of the options you have for drawing graphics on Android and which tasks they're best suited for.

Canvas and Drawables

Android provides a set of [View](#) widgets that provide general functionality for a wide array of user interfaces. You can also extend these widgets to modify the way they look or behave. In addition, you can do your own custom 2D rendering using the various drawing methods contained in the [Canvas](#) class or create [Drawable](#) objects for things such as textured buttons or frame-by-frame animations. **Hardware Acceleration**

Beginning in Android 3.0, you can hardware accelerate the majority of the drawing done by the Canvas APIs to further increase their performance.

OpenGL

Android supports OpenGL ES 1.0 and 2.0, with Android framework APIs as well as natively with the Native Development Kit (NDK). Using the framework APIs is desirable when you want to add a few graphical enhancements to your application that are not supported with the Canvas APIs, or if you desire platform independence and don't demand high performance. There is a performance hit in using the framework APIs compared to the NDK, so for many graphic intensive applications such as games, using the NDK is beneficial (It is important to note though that you can still get adequate performance using the framework APIs. For example, the Google Body app is developed entirely using the framework APIs). OpenGL with the NDK is also useful if you have a lot of native code that you want to port over to Android. For more information about using the NDK, read the docs in the [docs/](#) directory of the [NDK download](#).

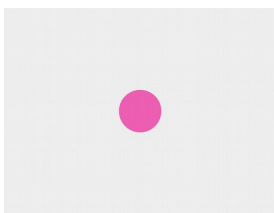
Gesture

Gestures are divided into Touch Mechanics (what your fingers do on the screen) and Touch Activities (results of specific gestures).

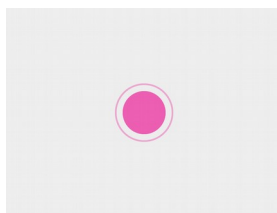
A touch mechanic (such as swipe) may have multiple results (such as enable/disable), depending on the context. A touch activity (zoom in) may be achieved through multiple touch mechanics (**pinch open**, **double touch**, etc.).

Touch mechanics

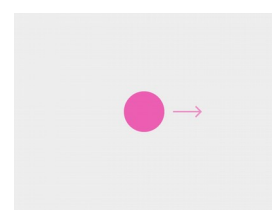
Touch mechanics refer to what the user's fingers do on the screen.



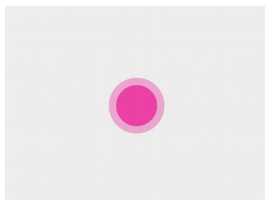
Touch



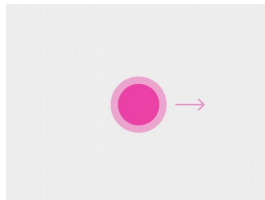
Double Touch



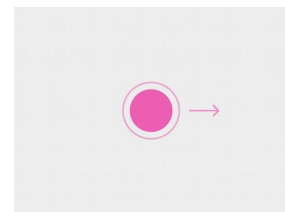
Fling, Drag, Swipe



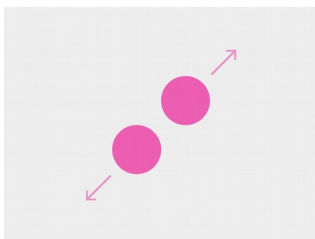
Long Press



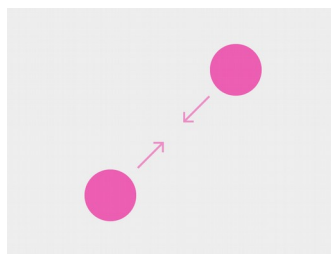
Long Press Drag



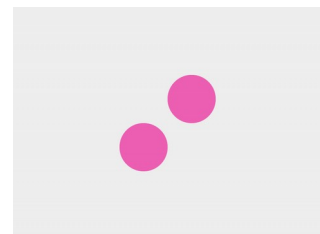
Double Touch Drag



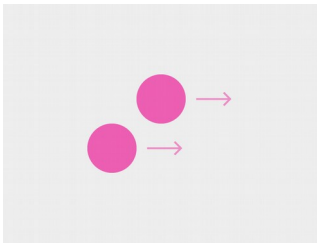
Pinch Open



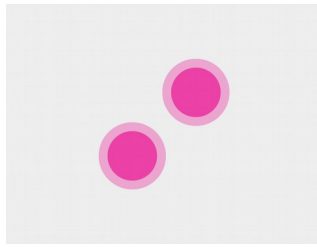
Pinch close



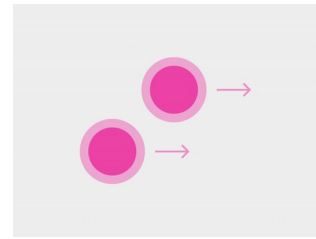
Two finger touch



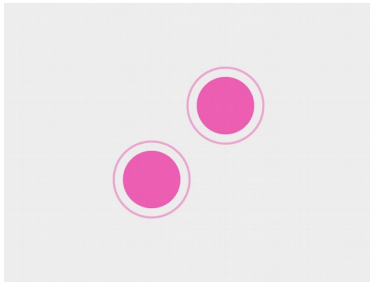
Two finger, swipe, drag,
Fling



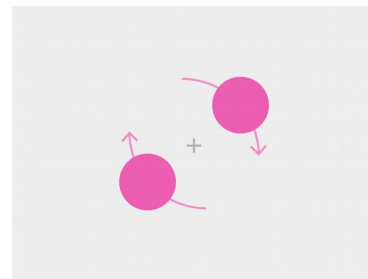
Two finger long press



Two finger long press drag



Two finger double touch



Rotate

Action	Description	Touch mechanics
Activate	Activates a screen element, like a button	Touch
Cancel or Escape	Cancels or escapes out of the current task, as in dialogs or menus	Touch
Enable/Disable lights out	Hides or shows a view's chrome	Touch
Drag or Swipe or Fling	See the following section Drag, swipe, or fling details for distinctions between Scroll, Reveal upon scroll, Pan, Dismiss, Swipe to refresh, Edge swipe, Paging swipe, Overscroll collapse, Menu open, and Tilt	Drag, Swipe, or Fling
Data selection (when nothing is selected)	Selects a single element	Long press, two-finger touch
Data selection (when items are already selected)	Selects additional elements while in selection mode Can use any combination of subsequent one- or two-finger gestures	Touch, two-finger touch
Data multi-selection drag	Reveals selection box that originates from point of gesture initiation The height and width can be adjusted based on finger position. The final selection is based on selection box dimensions upon finger(s) lifting.	Two-finger swipe or drag, long-press drag with no items selected
Pick up and move	Affects the selected item or items. It can be used to:	Two-finger long-press drag, long-press drag on selected item

	• Rearrange data within a view • Move an item into a container or onto a target • Reorder items in a list or a card collection	
Zoom in	Scales up content	• Double-touch • Double-touch drag (down) • Pinch open
Zoom in to fit	For nested views, scales up the smallest targetable view	Double-touch
Zoom out	Scales down content	• Double-touch at maximum zoom • Double-touch drag (up) • Pinch closed • Two-finger touch • Two-finger double touch
Expand	Expands collapsed content	Pinch open
Collapse	Collapses expanded content	Pinch closed
Rotate	Rotates the targeted content	Rotate

Multimedia

Multimedia is the field concerned with the computer controlled integration of text, graphics, drawings, still and moving images (Video), animation, audio, and any other media where every type of information can be represented, stored, transmitted and processed digitally.

Multimedia Applications

Examples of Multimedia Applications include:

- World Wide Web
- Multimedia Authoring, e.g. Adobe/Macromedia Director
- Video-on-demand
- Interactive TV
- Computer Games
- Virtual reality
- Digital video editing and production systems
- Multimedia Database systems

Multimedia Systems

A Multimedia System is a system capable of processing multimedia data and applications. A Multimedia System is characterised by the processing, storage, generation, manipulation and rendition of Multimedia information.